ASSIGNMENT-01

Implementation of all types of inheritance

Steps for Execution:

1. Create a folder named InheritanceDemo.
2. Inside it, create a folder named record.
3. Open the InheritanceDemo folder in VS Code/Eclipse.
4. Compile and execute the Java program.
5. If there are no compilation errors, the program will execute successfully.
6. The Single Inheritance operation will be performed first.
7. The parent class method and child class method will be displayed.
8. The Multilevel Inheritance operation will be performed next.
9. The grandparent, parent, and child class methods will be displayed.
10. The Hierarchical Inheritance operation will be performed next.
11. The parent class method and methods of both child classes will be displayed.
12. The Hybrid Inheritance operation will be performed next.
13. The methods inherited through multilevel and hierarchical inheritance will be displayed.
14. The output of all four types of inheritance will be displayed.
15. The program execution is completed.

Algorithm: **All Types of Inheritance**

Step 1: Start the program.

Step 2: Create the parent class Animal1 with the eat() method.

Step 3: Create the child class Dog1 extending Animal1 with the bark() method.

Step 4: Create an object of Dog1 and call eat() and bark() to demonstrate Single Inheritance.

Step 5: Create the parent class Animal2 with the eat() method.

Step 6: Create Dog2 extending Animal2 with the bark() method.

Step 7: Create Puppy2 extending Dog2 with the play() method.

Step 8: Create an object of Puppy2 and call eat(), bark(), and play() to demonstrate Multilevel Inheritance.

Step 9: Create the parent class Animal3 with the eat() method.

Step 10: Create two child classes Dog3 and Cat3 extending Animal3.

Step 11: Define bark() in Dog3 and meow() in Cat3.

Step 12: Create objects of Dog3 and Cat3 and call their methods to demonstrate Hierarchical Inheritance.

Step 13: Create the parent class Animal4 with the eat() method.

Step 14: Create Dog4 and Cat4 extending Animal4.

Step 15: Create Puppy4 extending Dog4 with the play() method.

Step 16: Create objects of Dog4, Cat4, and Puppy4 and call their respective methods.

Step 17: The combination of Multilevel Inheritance (Animal → Dog → Puppy) and Hierarchical Inheritance (Animal → Dog and Animal → Cat) demonstrates Hybrid Inheritance.

Step 18: Display the output of all four types of inheritance.

Program:

```
package assignment;
//Single Inheritance(Animal->Dog)
//One parent class->One child class
class Animal1 {
//Method for parent class
    void eat() {
        System.out.println("Animal eats food");
    }
}
class Dog1 extends Animal1 {
//Method for Dog1
   void bark() {
       System.out.println("Dog barks");
  }
}
//Multilevel Inheritance(Animal->Dog->Puppy)
//One  class inherits another class
//Grandparent class
class Animal2 {
  void eat() {
      System.out.println("Animal eats food");
  }
}
//Dog2 inherits Animal2
class Dog2 extends Animal2 {
   void bark() {
      System.out.println("Dog barks");
 }
}
//Puppy2 inherits Dog2
class Puppy2 extends Dog2 {
```

```
    void play() {
        System.out.println("Puppy plays");
  }
}
//Hierarchial Inheritance(   Animal   )
//                    /  \
//                  Dog    Cat
class Animal3 {
    void eat() {
        System.out.println("Animal eats food");
   }
}
//Dog3 inherits Animal3
class Dog3 extends Animal3 {
    void bark() {
        System.out.println("Dog barks");
  }
}
//Cat3 inherits Animal3
class Cat3 extends Animal3 {
    void meow() {
        System.out.println("Cat meows");
  }
}
//Hybrid Inheritance(Multilevel+Hierarchial)
//Hierarchial:Animal4->Dog4,Animal4->Cat4
//Multilevel:Animal4->Dog4->Puppy4
class Animal4 {
     void eat() {
        System.out.println("Animal eats food");
 }
}
//Dog4 inherits Animal4
class Dog4 extends Animal4 {
    void bark() {
```

```
        System.out.println("Dog barks");
 }
}
//Cat4 inherits Animal4
class Cat4 extends Animal4 {
    void meow() {
        System.out.println("Cat meows");
 }
}
//Puppy4 inherits Dog4
class Puppy4 extends Dog4 {
    void play() {
        System.out.println("Puppy plays");
 }
}
//Main Class
public class TypesOfInheritance {
     public static void main(String[] args) {
// Single Inheritance
        System.out.println("Single Inheritance");
//Create an object for Dog1
        Dog1 d1 = new Dog1();
//eat() comes from Animal1
       d1.eat();
//bark() comes from Dog1
       d1.bark();
// Multilevel Inhertitance
       System.out.println("Multilevel Inhertitance");
//Create an object for Puppy2
       Puppy2 p2 = new Puppy2();
//eat() comes from Animal2
       p2.eat();
//bark() comes from Dog2
       p2.bark();
//play() belongs to Puppy2
```

```
    p2.play();
//Hierarchial Inheritance
    System.out.println("Hierarchial Inheritance");
//Create an object for Dog3
    Dog3 d3 = new Dog3();
//eat() comes from Animal3
    d3.eat();
//bark() belongs to Dog3
    d3.bark();
//Create an object for Cat3
    Cat3 c3 = new Cat3();
//eat() comes from Animal3
    c3.eat();
//meow() belongs to Cat3
    c3.meow();
//Hybrid Inheritance
    System.out.println("Hybrid Inheritance");
//Create an object for Dog4
    Dog4 d4 = new Dog4();
//eat() comes from Animal4
    d4.eat();
//bark() belongs to Dog4
    d4.bark();
 //Create an object for Cat4
    Cat4 c4 = new Cat4();
//eat() comes from Animal4
    c4.eat();
//meow() belongs to Cat4
    c4.meow();
//Create an object for Puppy4
    Puppy4 p4 = new Puppy4();
//eat() comes from Animal4
    p4.eat();
//bark() comes from Dog4
    p4.bark();
```

```
//play() belongs to Puppy4
 p4.play();
 }
}
//Output:
//Single Inheritance: Parent method:Animal eats
//Single Inheritance: Child method:Dog barks
//Multilevel Inheritance: Grandparent method:Animal eats
//Multilevel Inheritance: Parent method:Dog barks
//Multilevel Inheritance: Child method:Puppy plays
//Hierarchical Inheritance: Parent method:Animal eats
//Hierarchical Inheritance: Child1 method:Dog barks
//Hierarchical Inheritance: Parent method:Animal eats
//Hierarchical Inheritance: Child2 method:Cat meows
//Hybrid Inheritance:Animal eats
//Hybrid Inheritance:Dog barks
//Hybrid Inheritance:Puppy plays
//Hybrid Inheritance:Animal eats
//Hybrid Inheritance:Cat meows
```

Output Screenshot: